

Affine vertex operator superalgebras at admissible levels

Simon Wood

School of Mathematics
Cardiff University

Joint work with David Ridout and John Snadden

Affine $\mathfrak{osp}(1|2)$ at level $-\frac{5}{4}$

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Outline

- 1 Finite $\mathfrak{osp}(1|2)$
- 2 Affine $\mathfrak{osp}(1|2)$
- 3 Modular properties and Verlinde formulae

Finite $\mathfrak{osp}(1|2)$

$$\mathfrak{osp}(1|2) = \text{span}\{e, x, h, y, f\} \supset \text{span}\{e, h, f\} = \mathfrak{sl}(2)$$

Commutation relations:

$$[h, e] = 2e, \quad [h, f] = -2f, \quad [e, f] = h$$

Normalised inner product

$$\kappa(h, h) = 2, \quad \kappa(e, f) = \kappa(f, e) = 1, \quad \kappa(x, y) = -\kappa(y, x) = 2$$

$\mathfrak{osp}(1|2)$ simple root $\alpha \leftrightarrow x$

$\mathfrak{sl}(2)$ simple root $2\alpha \leftrightarrow e$

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Commutation relations:

$$\begin{aligned} [h, x] &= x, & [h, y] &= -y, & [e, x] &= [f, y] = 0, \\ [e, y] &= -x, & [f, x] &= -y \end{aligned}$$

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Simple $\mathfrak{sl}(2)$ and $\mathfrak{osp}(1|2)$ weight modules

Cartan subalgebra centraliser: $\text{Com}(\mathfrak{h}, \mathfrak{sl}(2)) = \mathbb{C}[h, Q]$, where $Q = \frac{1}{2}h^2 + ef + fe$

Theorem [Mazorchuk]

Every simple $\mathfrak{sl}(2)$ weight module is isomorphic to one of the following:

- $\overline{\mathcal{F}}_\lambda$, $\lambda \in \mathbb{Z}_{\geq 0}$: Finite-dimensional module of highest weight $\lambda\alpha$.
- $\overline{\mathcal{D}}_\lambda^+$, $\lambda \in \mathbb{C} \setminus \mathbb{Z}_{\geq 0}$: Infinite-dimensional highest weight module of weight $\lambda\alpha$.
- $\overline{\mathcal{D}}_\lambda^-$, $\lambda \in \mathbb{C} \setminus \mathbb{Z}_{\leq 0}$: Infinite-dimensional lowest weight module of weight $\lambda\alpha$.
- $\overline{\mathcal{E}}_{\Lambda, q}$, $q \in \mathbb{C}$, $\Lambda \in \mathbb{C}/2\mathbb{Z}$ and $q \neq \frac{1}{2}\lambda(\lambda + 2)$ for all $\lambda \in \Lambda$: Dense module, $Q = q \cdot \text{id}$.

Simple $\mathfrak{sl}(2)$ and $\mathfrak{osp}(1|2)$ weight modules

Cartan subalgebra centraliser: $\text{Com}(\mathfrak{h}, \mathfrak{osp}(1|2)) = \mathbb{C}[h, \Sigma]$, where $\Sigma = \frac{1}{2} + xy - yx$, $Q^{\mathfrak{osp}} = \frac{1}{2}h^2 + ef + fe - \frac{1}{2}xy + \frac{1}{2}yx = \frac{1}{2}\Sigma^2 - \frac{1}{8}$

Theorem [Snadden-Ridout-SW]

Every simple $\mathfrak{osp}(1|2)$ weight module is isomorphic to one of the following (or their parity reversals):

- $\overline{\mathcal{A}}_\lambda$, $\lambda \in \mathbb{Z}_{\geq 0}$: Finite-dimensional module of highest weight $\lambda\alpha$.
- $\overline{\mathcal{B}}_\lambda^+$, $\lambda \in \mathbb{C} \setminus \mathbb{Z}_{\geq 0}$: Infinite-dimensional highest weight module of weight $\lambda\alpha$.
- $\overline{\mathcal{B}}_\lambda^-$, $\lambda \in \mathbb{C} \setminus \mathbb{Z}_{\leq 0}$: Infinite-dimensional lowest weight module of weight $\lambda\alpha$.
- $\overline{\mathcal{C}}_{\Lambda, s}$, $s \in \mathbb{C}$, $\Lambda \in \mathbb{C}/2\mathbb{Z}$ and $\lambda^2 \neq (s - \frac{1}{2})^2$ for all $\lambda \in \Lambda$: Dense module, $\Sigma = s \cdot \text{id}$.

Affine $\mathfrak{osp}(1|2)$

NS algebra:

$$\widehat{\mathfrak{osp}(1|2)}_{NS} = \text{span}\{h_m, e_m, f_m : m \in \mathbb{Z}\} \cup \{x_m, y_m : m \in \mathbb{Z}\} \cup \{K, L_0\}$$

R algebra:

$$\widehat{\mathfrak{osp}(1|2)}_R = \text{span}\{h_m, e_m, f_m : m \in \mathbb{Z}\} \cup \{x_m, y_m : m \in \mathbb{Z} + \tfrac{1}{2}\} \cup \{K, L_0\}$$

Decomposition: $\widehat{\mathfrak{osp}(1|2)} = \mathfrak{n} \oplus \mathfrak{z} \oplus \mathfrak{p}$

$\mathfrak{n}$: negative modes, $\mathfrak{p}$: positive modes, $\mathfrak{z}$: zero modes

Definition

Let $\mathcal{R}$ be the full subcategory of $\mathfrak{osp}(1|2) - \text{Mod}$ whose objects M satisfy:

- 1 M is finitely generated.
- 2 M is a weight module (Cartan acts semisimply).
- 3 Action of $\mathfrak{p}$ is locally nilpotent (for any $m \in M$, $U(\mathfrak{p})m$ is finite-dimensional).

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Induced modules

NS sector: M simple quotient of $U(\widehat{\mathfrak{osp}(1|2)}_{NS}) \otimes_{U(\mathfrak{z} \oplus \mathfrak{p})} \overline{M}$,
 $\overline{M}$ simple $\mathfrak{osp}(1|2)$ weight module and $K = k \cdot \text{id}$.

R sector: N simple quotient of $U(\widehat{\mathfrak{osp}(1|2)}_R) \otimes_{U(\mathfrak{z} \oplus \mathfrak{p})} \overline{N}$,
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Universal $\mathfrak{osp}(1|2)$ VOSA: $V^k(\mathfrak{osp}(1|2)) = U(\widehat{\mathfrak{osp}(1|2)}_{NS}) \otimes_{U(\mathfrak{z} \oplus \mathfrak{p})} \overline{\mathcal{A}}_0$

Proposition [Kac-Gorelik]

Universal $\mathfrak{osp}(1|2)$ VOSA is reducible if and only if k is critical or if there exist $u, v \in \mathbb{Z}$, $u \geq 2$, $v \geq 1$, $u - v \in 2\mathbb{Z}$, $\gcd\{u, \frac{u-v}{2}\} = 1$ such that $2k + 3 = \frac{u}{v}$.

Simple $\mathfrak{osp}(1|2)$ VOSA: $L_k(\mathfrak{osp}(1|2))$.

Today: fix $k = -\frac{5}{4}$, $u = 2$, $v = 4$.

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Simple $\mathfrak{osp}(1|2)$ at level $-\frac{5}{4}$

Singular vector: $\chi = (x_{-2} - 4y_{-1}e_1 + 2h_{-1}x_{-1})\Omega_{-\frac{5}{4}}$

Weight 0 descendant: $\phi = y_0\chi = (h_{-2} + 2h_{-1}^2 + 8f_{-1}e_{-1} + 6y_{-1}x_{-1})\Omega_{-\frac{5}{4}}$

Theorem [Snadden-Ridout-SW], [Frenkel-Zhu], [Kac-Wang], [Dong-Li-Mason]

- 1 Universal Zhu algebras:

$$\text{Zhu}[V^k(\mathfrak{osp}(1|2))] = U(\mathfrak{osp}(1|2)), \quad \text{Zhu}^\tau[V^k(\mathfrak{osp}(1|2))] = U(\mathfrak{sl}(2))$$

- 2 Simple quotient Zhu algebras, k admissible:

$$\text{Zhu}[L_k(\mathfrak{osp}(1|2))] = \frac{U(\mathfrak{osp}(1|2))}{\langle [\phi] \rangle}, \quad \text{Zhu}^\tau[L_k(\mathfrak{osp}(1|2))] = \frac{U(\mathfrak{sl}(2))}{\langle [\phi]^\tau \rangle}$$

- 3 Singular vector images at level $k = -\frac{5}{4}$:

$$[\phi] = \Sigma(2\Sigma - 1), \quad [\phi]^\tau = 4Q + \frac{15}{8} = 4(Q + \frac{15}{32})$$

Simple module classification

Theorem

Any simple $L_{-\frac{5}{4}}(\mathfrak{osp}(1|2))$ module in $\mathcal{R}$ is isomorphic to one of the following or its parity reversal:

NS sector: $\mathcal{A}_0, \mathcal{B}_{-\frac{1}{2}}^+, \mathcal{B}_{\frac{1}{2}}^-, \mathcal{C}_{\Lambda,0}, \Lambda \in \mathbb{C}/2\mathbb{Z} \setminus [\pm\frac{1}{2}]$.

R sector: $\mathcal{D}_{-\frac{5}{4}}^+, \mathcal{D}_{-\frac{3}{4}}^+, \mathcal{D}_{\frac{5}{4}}^-, \mathcal{D}_{\frac{3}{4}}^-, \mathcal{E}_{\Lambda, -\frac{15}{32}}, \Lambda \in \mathbb{C}/2\mathbb{Z} \setminus [\pm\frac{5}{4}]$.

Kac table:

	0	1	2	3
1	$\mathcal{A}_0$	$\mathcal{D}_{-\frac{3}{4}}^+$ $\mathcal{D}_{\frac{5}{4}}^-$ $\mathcal{E}_{\Lambda, -\frac{15}{32}}$	$\mathcal{B}_{-\frac{1}{2}}^+$ $\mathcal{B}_{\frac{1}{2}}^-$ $\mathcal{C}_{\Lambda,0}$	$\mathcal{D}_{-\frac{5}{4}}^+$ $\mathcal{D}_{\frac{3}{4}}^-$ $\mathcal{E}_{\Lambda, -\frac{15}{32}}$

Algebra automorphisms

Conjugation c :

$$\begin{array}{lll} ce_n = -f_n & ch_n = -h_n & cf_n = -e_n \\ cx_n = -y_n & cK = K & cy_n = x_n \\ & cL_0 = L_0 & \end{array}$$

Spectral flow σ :

$$\begin{array}{lll} \sigma^\ell e_n = e_{n-2\ell} & \sigma^\ell h_n = h_n - 2\ell\delta_{n,0}K & \sigma^\ell f_n = f_{n+2\ell} \\ \sigma^\ell x_n = x_{n-\ell} & \sigma^\ell K = K & \sigma^\ell y_n = y_{n+\ell} \\ \sigma^\ell L_0 = L_0 - \ell h_0 + \ell^2 K & & \end{array}$$

σ^ℓ , $\ell \in \mathbb{Z} + \frac{1}{2}$, is an isomorphism of the NS and R algebras

Let $\alpha = c, \sigma^\ell$ and M a module, then $\alpha M \cong M$ and for $m \in M$,
 $g \cdot \alpha m = \alpha^{-1}(g)m$, $g \in \widehat{\mathfrak{osp}(1|2)}$.

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Automorphism twist coincidences

Definition

Let $\mathcal{T}$ be the full subcategory of $L_{-\frac{5}{4}}(\mathfrak{osp}(1|2)) - \mathbf{Mod}$ generated by $\mathcal{R}$, its twists by c and σ^ℓ , and (finite) extensions between different twists of $\mathcal{R}$.

$$\begin{aligned} c\mathcal{A}_0 &\cong \mathcal{A}_0, & c\mathcal{B}_{-\frac{1}{2}}^+ &\cong \mathcal{B}_{\frac{1}{2}}^-, & c\mathcal{C}_\lambda &\cong \mathcal{C}_{-\lambda} \\ c\mathcal{D}_{-\frac{3}{4}}^+ &\cong \mathcal{D}_{\frac{3}{4}}^-, & c\mathcal{D}_{-\frac{5}{4}}^+ &\cong \mathcal{D}_{\frac{5}{4}}^-, & c\mathcal{E}_\lambda &\cong \mathcal{E}_{-\lambda} \\ \sigma^{\frac{1}{2}}\mathcal{A}_0 &\cong \mathcal{D}_{-\frac{5}{4}}^+, & \sigma^{-\frac{1}{2}}\mathcal{A}_0 &\cong \mathcal{D}_{\frac{5}{4}}^-, & \sigma^{-\frac{1}{2}}\mathcal{B}_{-\frac{1}{2}}^+ &\cong \mathcal{D}_{\frac{3}{4}}^-, & \sigma^{\frac{1}{2}}\mathcal{B}_{\frac{1}{2}}^- &\cong \mathcal{D}_{-\frac{3}{4}}^+. \end{aligned}$$

Theorem

Any simple module M in $\mathcal{T}$ is isomorphic to one of the following or its parity reversal:

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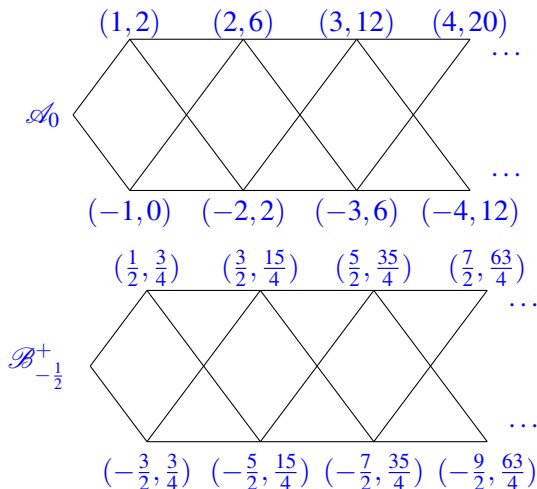
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Any simple module M in $\mathcal{T}$ is isomorphic to one of the following or its parity reversal:

$$\sigma^\ell \mathcal{A}_0, \quad \sigma^\ell \mathcal{B}_{-\frac{1}{2}}^+, \quad \sigma^\ell \mathcal{B}_{\frac{1}{2}}^-, \quad \sigma^\ell \mathcal{C}_\lambda, \quad \sigma^\ell \mathcal{E}_\mu,$$

where $\ell \in \frac{1}{2}\mathbb{Z}$, $\lambda \neq \frac{1}{2} \pmod{1}$, $\mu \neq \pm\frac{5}{4} \pmod{2}$.

Verma module embeddings [Ihara-Koga]



Characters and supercharacters

$$\mathrm{ch}_0[\mathcal{A}_0] = y^{-\frac{5}{4}} q^{\frac{5}{24}} \prod_{i=1}^{\infty} \frac{(1 - zq^{2i})(1 + zq^i)(1 + q^i)(1 + z^{-1}q^i)(1 - z^{-1}q^{2i})}{(1 - z^2q^i)(1 - z^{-2}q^i)}$$

$$\mathrm{ch}_{\frac{1}{2}}[\mathcal{A}_0] = y^{-\frac{5}{4}} q^{\frac{5}{24}} \prod_{i=1}^{\infty} \frac{(1 + zq^{2i})(1 - zq^i)(1 + q^i)(1 - z^{-1}q^i)(1 + z^{-1}q^{2i})}{(1 - z^2q^i)(1 - z^{-2}q^i)}$$

$$|q| < 1, \quad |q|^{\frac{1}{2}} < |z| < |q|^{-\frac{1}{2}}$$

$$\mathrm{ch}_0\left[\mathcal{B}_{-\frac{1}{2}}^+\right] = y^{-\frac{5}{4}} z^{-\frac{1}{2}} q^{\frac{1}{24}} \prod_{i=1}^{\infty} \frac{(1 - zq^{2i-1})(1 + zq^i)(1 - q^{2i})(1 + z^{-1}q^{i-1})(1 - z^{-1}q^{2i-1})}{(1 - z^2q^i)(1 - z^{-2}q^{i-1})}$$

$$\mathrm{ch}_{\frac{1}{2}}\left[\mathcal{B}_{-\frac{1}{2}}^+\right] = y^{-\frac{5}{4}} z^{-\frac{1}{2}} q^{\frac{1}{24}} \prod_{i=1}^{\infty} \frac{(1 + zq^{2i-1})(1 - zq^i)(1 - q^{2i})(1 - z^{-1}q^{i-1})(1 + z^{-1}q^{2i-1})}{(1 - z^2q^i)(1 - z^{-2}q^{i-1})}$$

$$|q| < 1, \quad 1 < |z| < |q|^{-1}$$

Remaining highest and lowest weight characters follow from automorphism twists.

Characters and supercharacters

$$\mathrm{ch}_0[\mathcal{A}_0] = y^{-\frac{5}{4}} q^{\frac{5}{24}} \prod_{i=1}^{\infty} \frac{(1 - zq^{2i})(1 + zq^i)(1 + q^i)(1 + z^{-1}q^i)(1 - z^{-1}q^{2i})}{(1 - z^2q^i)(1 - z^{-2}q^i)}$$

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$$|q| < 1, \quad 1 < |z| < |q|^{-1}$$

Remaining highest and lowest weight characters follow from automorphism twists.

Characters and supercharacters

The exact sequences

$$0 \longrightarrow \Pi \mathcal{B}_{-\frac{1}{2}}^+ \longrightarrow \mathcal{C}_{\frac{1}{2}}^+ \longrightarrow \mathcal{B}_{\frac{1}{2}}^- \longrightarrow 0$$

$$0 \longrightarrow \mathcal{D}_{-\frac{5}{4}}^+ \longrightarrow \mathcal{E}_{\frac{3}{4}}^+ \longrightarrow \mathcal{D}_{\frac{3}{4}}^- \longrightarrow 0$$

$$0 \longrightarrow \mathcal{D}_{-\frac{3}{4}}^+ \longrightarrow \mathcal{E}_{\frac{5}{4}}^+ \longrightarrow \mathcal{D}_{\frac{5}{4}}^- \longrightarrow 0$$

imply character formulae for $\mathcal{C}_{\frac{1}{2}}^+$, $\mathcal{E}_{\frac{3}{4}}^+$ and $\mathcal{E}_{\frac{5}{4}}^+$.

Characters and supercharacters

Conjecture / soon to be Proposition

$$\begin{aligned}\mathrm{ch}_\varepsilon [\mathcal{C}_\lambda] &= \frac{y^{-\frac{5}{4}} z^\lambda}{\sqrt{2}} A_2(q) \sum_{n \in \mathbb{Z}} ((-1)^\varepsilon z)^n, \\ \mathrm{ch}_\varepsilon [\mathcal{E}_\mu] &= \frac{y^{-\frac{5}{4}} z^\mu}{2} \left(A_3(q) \sum_{n \in \mathbb{Z}} ((-1)^\varepsilon z)^n + A_4(q) \sum_{n \in \mathbb{Z}} ((-1)^{\varepsilon+1} z)^n \right),\end{aligned}$$

where the $A_i(q)^2$ are ratios of ϑ and η functions.

The (super)characters of the $\sigma^\ell \mathcal{C}_\lambda$ and $\sigma^\ell \mathcal{E}_\mu$ are linearly independent.

We can identify elements of the Grothendieck group with pairs of characters and supercharacters, that is, for M a module

$$[M] = \left(\mathrm{ch}_0 [M], \mathrm{ch}_{\frac{1}{2}} [M] \right)$$

Characters and supercharacters

Conjecture / soon to be Proposition

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We can identify elements of the Grothendieck group with pairs of characters and supercharacters, that is, for M a module

$$[M] = \left(\mathrm{ch}_0 [M], \mathrm{ch}_{\frac{1}{2}} [M] \right)$$

Resolving non-dense simples

Repeated splicing of

$$0 \longrightarrow \Pi \mathcal{B}_{-\frac{1}{2}}^+ \longrightarrow \mathcal{C}_{\frac{1}{2}}^+ \longrightarrow \mathcal{B}_{\frac{1}{2}}^- \longrightarrow 0$$

$$0 \longrightarrow \mathcal{D}_{-\frac{5}{4}}^+ \longrightarrow \mathcal{C}_{\frac{3}{4}}^+ \longrightarrow \mathcal{D}_{\frac{3}{4}}^- \longrightarrow 0$$

$$0 \longrightarrow \mathcal{D}_{-\frac{3}{4}}^+ \longrightarrow \mathcal{C}_{\frac{5}{4}}^+ \longrightarrow \mathcal{D}_{\frac{5}{4}}^- \longrightarrow 0$$

yields:

Proposition

$$[\mathcal{A}_0] = \sum_{n=0}^{\infty} \left(\left[\sigma^{4n+1/2} \mathcal{C}_{5/4}^+ \right] - \left[\sigma^{4n+1} \mathcal{C}_{1/2}^+ \right] + \left[\Pi \sigma^{4n+3/2} \mathcal{C}_{3/4}^+ \right] \right. \\ \left. - \left[\Pi \sigma^{4n+5/2} \mathcal{C}_{5/4}^+ \right] + \left[\Pi \sigma^{4n+3} \mathcal{C}_{1/2}^+ \right] - \left[\sigma^{4n+7/2} \mathcal{C}_{3/4}^+ \right] \right),$$

Resolving non-dense simples

Repeated splicing of

$$\begin{aligned}0 &\longrightarrow \Pi \mathcal{B}_{-\frac{1}{2}}^+ \longrightarrow \mathcal{C}_{\frac{1}{2}}^+ \longrightarrow \mathcal{B}_{\frac{1}{2}}^- \longrightarrow 0 \\0 &\longrightarrow \sigma^{\frac{1}{2}} \mathcal{A}_0 \longrightarrow \mathcal{C}_{\frac{3}{4}}^+ \longrightarrow \sigma^{-\frac{1}{2}} \mathcal{B}_{-\frac{1}{2}}^+ \longrightarrow 0 \\0 &\longrightarrow \sigma^{\frac{1}{2}} \mathcal{B}_{\frac{1}{2}}^- \longrightarrow \mathcal{C}_{\frac{5}{4}}^+ \longrightarrow \sigma^{-\frac{1}{2}} \mathcal{A}_0 \longrightarrow 0\end{aligned}$$

yields:

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$$\begin{aligned}[\mathcal{A}_0] = \sum_{n=0}^{\infty} \bigg(& \left[\sigma^{4n+1/2} \mathcal{C}_{5/4}^+ \right] - \left[\sigma^{4n+1} \mathcal{C}_{1/2}^+ \right] + \left[\Pi \sigma^{4n+3/2} \mathcal{C}_{3/4}^+ \right] \\ & - \left[\Pi \sigma^{4n+5/2} \mathcal{C}_{5/4}^+ \right] + \left[\Pi \sigma^{4n+3} \mathcal{C}_{1/2}^+ \right] - \left[\sigma^{4n+7/2} \mathcal{C}_{3/4}^+ \right] \bigg),\end{aligned}$$

Resolving non-dense simples

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Resolving non-dense simples

Repeated splicing of

$$\begin{aligned}0 &\longrightarrow \Pi \mathcal{B}_{-\frac{1}{2}}^+ \longrightarrow \mathcal{C}_{\frac{1}{2}}^+ \longrightarrow \mathcal{B}_{\frac{1}{2}}^- \longrightarrow 0 \\0 &\longrightarrow \sigma^{\frac{1}{2}} \mathcal{A}_0 \longrightarrow \mathcal{C}_{\frac{3}{4}}^+ \longrightarrow \sigma^{-\frac{1}{2}} \mathcal{B}_{-\frac{1}{2}}^+ \longrightarrow 0 \\0 &\longrightarrow \sigma^{\frac{1}{2}} \mathcal{B}_{\frac{1}{2}}^- \longrightarrow \mathcal{C}_{\frac{5}{4}}^+ \longrightarrow \sigma^{-\frac{1}{2}} \mathcal{A}_0 \longrightarrow 0\end{aligned}$$

yields:

Proposition

$$\begin{aligned}\left[\mathcal{B}_{1/2}^-\right] &= \sum_{n=0}^{\infty} \left(\left[\sigma^{4n} \mathcal{C}_{1/2}^+\right] - \left[\Pi \sigma^{4n+1/2} \mathcal{C}_{3/4}^+\right] + \left[\Pi \sigma^{4n+3/2} \mathcal{C}_{5/4}^+\right] \right. \\&\quad \left. - \left[\Pi \sigma^{4n+2} \mathcal{C}_{1/2}^+\right] + \left[\sigma^{4n+5/2} \mathcal{C}_{3/4}^+\right] - \left[\sigma^{4n+7/2} \mathcal{C}_{5/4}^+\right] \right),\end{aligned}$$

Resolving non-dense simples

Repeated splicing of

$$\begin{aligned}0 &\longrightarrow \Pi \mathcal{B}_{-\frac{1}{2}}^+ \longrightarrow \mathcal{C}_{\frac{1}{2}}^+ \longrightarrow \mathcal{B}_{\frac{1}{2}}^- \longrightarrow 0 \\0 &\longrightarrow \sigma^{\frac{1}{2}} \mathcal{A}_0 \longrightarrow \mathcal{C}_{\frac{3}{4}}^+ \longrightarrow \sigma^{-\frac{1}{2}} \mathcal{B}_{-\frac{1}{2}}^+ \longrightarrow 0 \\0 &\longrightarrow \sigma^{\frac{1}{2}} \mathcal{B}_{\frac{1}{2}}^- \longrightarrow \mathcal{C}_{\frac{5}{4}}^+ \longrightarrow \sigma^{-\frac{1}{2}} \mathcal{A}_0 \longrightarrow 0\end{aligned}$$

yields:

Proposition

$$\begin{aligned}\left[\mathcal{B}_{-1/2}^+\right] &= \sum_{n=0}^{\infty} \left(\left[\sigma^{4n+1/2} \mathcal{C}_{3/4}^+\right] - \left[\sigma^{4n+3/2} \mathcal{C}_{5/4}^+\right] + \left[\sigma^{4n+2} \mathcal{C}_{1/2}^+\right] \right. \\&\quad \left. - \left[\Pi \sigma^{4n+5/2} \mathcal{C}_{3/4}^+\right] + \left[\Pi \sigma^{4n+7/2} \mathcal{C}_{5/4}^+\right] - \left[\Pi \sigma^{4n+4} \mathcal{C}_{1/2}^+\right] \right).\end{aligned}$$

Modular transformations

Proposition

For $\ell \in \frac{1}{2}\mathbb{Z}$ and $\lambda \in \mathbb{R}$, let

$$K_\ell^T(\lambda) = e^{2\pi i \ell(k\ell + \lambda)}.$$

Then,

$$\begin{aligned} T \left\{ \text{ch}_\varepsilon \left[\sigma^\ell \mathcal{C}_\lambda \right] \right\} &= e^{\frac{\pi i}{12}} K_\ell^T(\lambda) \text{ch}_{\varepsilon+\ell} \left[\sigma^\ell \mathcal{C}_\lambda \right] \\ T \left\{ \text{ch}_\varepsilon \left[\sigma^\ell \mathcal{E}_\lambda \right] \right\} &= e^{-\frac{\pi i}{24}} K_\ell^T(\lambda) \text{ch}_{\varepsilon+\ell+\frac{1}{2}} \left[\sigma^\ell \mathcal{E}_\lambda \right] \end{aligned}$$

Modular transformations

Proposition

For $\ell, \ell' \in \frac{1}{2}\mathbb{Z}$ and $\lambda, \lambda' \in \mathbb{R}$, let

$$K_{\ell, \ell'}^S(\lambda, \lambda') = e^{-2\pi i(2k\ell\ell' + \ell'\lambda + \ell\lambda')}.$$

Then,

$$\begin{aligned} S\left\{\text{ch}_\varepsilon\left[\sigma^\ell \mathcal{E}_\lambda\right]\right\} &= \frac{1}{\sqrt{2}} \sum_{\ell' \in \mathbb{Z} + \varepsilon} \int_{\mathbb{R}/2\mathbb{Z}} K_{\ell, \ell'}^S(\lambda, \lambda') \text{ch}_{\ell + \frac{1}{2}}\left[\sigma^{\ell'} \mathcal{E}_{\lambda'}\right] d\lambda' \\ S\left\{\text{ch}_\varepsilon\left[\sigma^\ell \mathcal{E}_\lambda\right]\right\} &= \frac{1}{\sqrt{2}} \sum_{\ell' \in \mathbb{Z} + \varepsilon + \frac{1}{2}} \int_{\mathbb{R}/\mathbb{Z}} K_{\ell, \ell'}^S(\lambda, \lambda') \text{ch}_{\ell + \frac{1}{2}}\left[\sigma^{\ell'} \mathcal{E}_{\lambda'}\right] d\lambda' \\ &\quad + \frac{1}{2} \sum_{\ell' \in \mathbb{Z} + \varepsilon} \int_{\mathbb{R}/2\mathbb{Z}} K_{\ell, \ell'}^S(\lambda, \lambda') \text{ch}_{\ell + \frac{1}{2}}\left[\sigma^{\ell'} \mathcal{E}_{\lambda'}\right] d\lambda' \end{aligned}$$

S matrices

Recall: $K_{\ell,\ell'}^S(\lambda, \lambda') = e^{-2\pi i(2k\ell\ell' + \ell'\lambda + \ell\lambda')}$,

$$S_{\varepsilon,\varepsilon'} \left[\sigma^\ell \mathcal{C}_\lambda \rightarrow \sigma^{\ell'} \mathcal{C}_{\lambda'} \right] = 0,$$

$$S_{\varepsilon,\varepsilon'} \left[\sigma^\ell \mathcal{C}_\lambda \rightarrow \sigma^{\ell'} \mathcal{C}_{\lambda'} \right] = \frac{1}{\sqrt{2}} K_{\ell,\ell'}^S(\lambda, \lambda') \mathbf{1}_{\ell \neq \varepsilon'} \mathbf{1}_{\ell' = \varepsilon},$$

$$S_{\varepsilon,\varepsilon'} \left[\sigma^\ell \mathcal{C}_\lambda \rightarrow \sigma^{\ell'} \mathcal{C}_{\lambda'} \right] = \frac{1}{\sqrt{2}} K_{\ell,\ell'}^S(\lambda, \lambda') \mathbf{1}_{\ell = \varepsilon'} \mathbf{1}_{\ell' \neq \varepsilon},$$

$$S_{\varepsilon,\varepsilon'} \left[\sigma^\ell \mathcal{C}_\lambda \rightarrow \sigma^{\ell'} \mathcal{C}_{\lambda'} \right] = \frac{1}{2} K_{\ell,\ell'}^S(\lambda, \lambda') \mathbf{1}_{\ell = \varepsilon'} \mathbf{1}_{\ell' = \varepsilon},$$

Thus,

$$S = S^t, \quad SS^\dagger = \mathbf{1}, \quad S^2 = C.$$

Induction from bosonic Verlinde formulae

Assumptions: VOSA V with even subVOA O

- 1 V is an order 2 simple current extension of O .
- 2 V admits a set of even parity standard modules $\mathcal{S} = \mathcal{S}_0 \cup \mathcal{S}_{\frac{1}{2}}$.
- 3 The fusion rules of V and O are compatible.
- 4 Fusion for V and O is exact.
- 5 The standard module formalism applies to O , that is, the Grothendieck fusion of O is given by

$$\text{ch}[W] \boxtimes_O \text{ch}[X] = \int_{\mathcal{S}^O} \begin{bmatrix} Y \\ W \ X \end{bmatrix} \text{ch}[Y] dY,$$

where

$$\begin{bmatrix} Y \\ W \ X \end{bmatrix} = \int_{\mathcal{S}^O} \frac{s[X \rightarrow Z]s[W \rightarrow Z]s[Y \rightarrow Z]^*}{s[O \rightarrow Z]} dZ$$

Induction from bosonic Verlinde formulae

Under the previous assumptions we have

Theorem

Let M, N be V modules and let ξ, ξ' be 0 or $\frac{1}{2}$ depending on whether M and N are NS or R modules. Then,

$$\mathrm{ch}_\varepsilon [M] \boxtimes \mathrm{ch}_\varepsilon [N] = \int_{\mathcal{S}_{\xi+\xi'}} \begin{bmatrix} P \\ M \ N \end{bmatrix}_\varepsilon \mathrm{ch}_\varepsilon [P] \mathrm{d}P,$$

where

$$\begin{bmatrix} P \\ M \ N \end{bmatrix}_\varepsilon = \int_{\mathcal{S}_{\varepsilon+\frac{1}{2}}} \frac{S_{\varepsilon, \xi+\frac{1}{2}}[M \rightarrow Q] S_{\varepsilon, \xi'+\frac{1}{2}}[N \rightarrow Q] S_{\varepsilon, \xi+\xi'+\frac{1}{2}}[P \rightarrow Q]^*}{S_{\varepsilon, \frac{1}{2}}[V \rightarrow Q]} \mathrm{d}Q.$$

$\mathfrak{osp}(1|2)$ Verlinde formula

Assuming $L_{-\frac{5}{4}}(\mathfrak{osp}(1|2))$ satisfies all the previous assumptions:

Proposition

Let M, N be $L_{-\frac{5}{4}}(\mathfrak{osp}(1|2))$ modules and let $m, n \in \frac{1}{2}\mathbb{Z}$, then

$$[\Pi M] \boxtimes [N] = \Pi([M] \boxtimes [N])$$

$$[cM] \boxtimes [cN] = c([M] \boxtimes [N])$$

$$[\sigma^m M] \boxtimes [\sigma^n N] = \sigma^{m+n}([M] \boxtimes [N])$$

$\mathfrak{osp}(1|2)$ Verlinde formula

Assuming $L_{-\frac{5}{4}}(\mathfrak{osp}(1|2))$ satisfies all the previous assumptions:

Proposition

$$\begin{aligned}\left[\mathcal{B}_{-\frac{1}{2}}^{+}\right] \boxtimes \left[\mathcal{B}_{-\frac{1}{2}}^{+}\right] &= \left[\Pi \sigma \mathcal{B}_{+\frac{1}{2}}^{-}\right] + \left[\sigma^{\frac{1}{2}} \mathcal{E}_{\frac{1}{4}}\right], \\ \left[\mathcal{B}_{-\frac{1}{2}}^{+}\right] \boxtimes \left[\mathcal{B}_{+\frac{1}{2}}^{-}\right] &= [\mathcal{A}_0] + [\mathcal{C}_0], \\ \left[\mathcal{B}_{-\frac{1}{2}}^{+}\right] \boxtimes [\mathcal{C}_{\lambda}] &= \left[\sigma^{\frac{1}{2}} \mathcal{E}_{\lambda-k-\frac{1}{2}}\right] + \left[\Pi \sigma^{\frac{1}{2}} \mathcal{E}_{\lambda-k+\frac{1}{2}}\right] + \left[\mathcal{C}_{\lambda-\frac{1}{2}}\right], \\ \left[\mathcal{B}_{-\frac{1}{2}}^{+}\right] \boxtimes [\mathcal{E}_{\lambda}] &= \left[\sigma^{\frac{1}{2}} \mathcal{C}_{\lambda-k-\frac{1}{2}}\right] + \left[\mathcal{E}_{\lambda-\frac{1}{2}}\right],\end{aligned}$$

$\mathfrak{osp}(1|2)$ Verlinde formula

Assuming $L_{-\frac{5}{4}}(\mathfrak{osp}(1|2))$ satisfies all the previous assumptions:

Proposition

$$\begin{aligned} [\mathcal{C}_\lambda] \boxtimes [\mathcal{C}_\mu] &= \left[\Pi \sigma^{\frac{1}{2}} \mathcal{E}_{\lambda+\mu-k-1} \right] + \left[\sigma^{\frac{1}{2}} \mathcal{E}_{\lambda+\mu-k} \right] + 2 \left[\mathcal{C}_{\lambda+\mu} \right] \\ &\quad + \left[\sigma^{-\frac{1}{2}} \mathcal{E}_{\lambda+\mu+k} \right] + \left[\Pi \sigma^{-\frac{1}{2}} \mathcal{E}_{\lambda+\mu+k+1} \right], \\ [\mathcal{C}_\lambda] \boxtimes [\mathcal{E}_\mu] &= \left[\sigma^{\frac{1}{2}} \mathcal{C}_{\lambda+\mu-k} \right] + \left[\mathcal{E}_{\lambda+\mu} \right] + \left[\Pi \mathcal{E}_{\lambda+\mu+1} \right] + \left[\sigma^{-\frac{1}{2}} \mathcal{C}_{\lambda+\mu+k} \right], \\ [\mathcal{E}_\lambda] \boxtimes [\mathcal{E}_\mu] &= \left[\sigma^{\frac{1}{2}} \mathcal{E}_{\lambda+\mu-k} \right] + \left[\mathcal{C}_{\lambda+\mu} \right] + \left[\sigma^{-\frac{1}{2}} \mathcal{E}_{\lambda+\mu+k} \right], \end{aligned}$$

Thank you!